Technical Architecture and Data Source Discovery for Project Fessi: A Comprehensive Analysis of the German Waste Management Information Landscape

1. Introduction: The Digital Fragmentation of German Waste Management

The conception of "Fessi"—a knowledge graph-driven chatbot designed to assist university students and staff in navigating the complexities of German waste disposal—addresses a significant gap in the current municipal service infrastructure. The German waste management sector, while technically advanced in terms of physical recycling rates and engineering, suffers from a profound digital fragmentation. This fragmentation is not merely a technical inconvenience but a direct consequence of the federal administrative structure mandated by the *Kreislaufwirtschaftsgesetz* (Circular Economy Act). Under this legal framework, the responsibility for waste disposal is devolved to the *öffentlich-rechtliche Entsorgungsträger* (public waste disposal authorities), comprising over 400 distinct districts (*Landkreise*) and independent cities (*kreisfreie Städte*).[1]

For a knowledge graph project like Fessi, this decentralization presents a formidable challenge. There is no centralized federal API that acts as a single source of truth for collection schedules, sorting rules, or facility locations. Instead, the data landscape is a mosaic of high-quality Open Data portals in metropolitan centers like Berlin and Munich, proprietary Software-as-a-Service (SaaS) silos in rural districts, and legacy static documents in others. The primary objective of this report is to deconstruct this landscape, identifying the specific data egress points—ranging from RESTful APIs to scrapeable HTML widgets—that will feed the Fessi ingestion pipeline.

The following analysis is based on an exhaustive review of technical documentation, reverse-engineered API specifications from the open-source community, and municipal waste datasets. It posits that while the administrative landscape is fragmented, the digital infrastructure is increasingly consolidated under a "SaaS Oligopoly" of white-label app providers. By targeting these providers—specifically Abfall+, JumoMind, and Müllmax—Fessi can achieve a scalable integration strategy that covers a vast majority of German university towns without necessitating custom connectors for every single municipality. Furthermore, the report establishes the critical role of the *Abfallverzeichnis-Verordnung* (AVV) as the

semantic backbone for the knowledge graph, ensuring that the chatbot's natural language processing (NLP) layer is grounded in legal classification standards.[2]

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2. The SaaS Oligopoly: Reverse-Engineering the Primary Data Sources

The most significant finding for the scalable development of Fessi is the prevalence of white-label mobile applications. A detailed examination of the digital footprint of various "Landkreis" waste apps reveals that they are rarely bespoke software solutions. Instead, they are localized skins of a few dominant backend platforms. Integrating with these platforms provides access to collection schedules (*Abfuhrkalender*) and local waste dictionaries (*Abfall-ABC*) for hundreds of municipalities simultaneously.

2.1 The Abfall+ / Abfall.io Ecosystem

The Abfall+ platform (also operating under the domains abfall.io and abfallplus.de) appears to be the market leader for small-to-medium-sized districts, particularly in southern and western Germany. The platform is characterized by a standardized, albeit undocumented, API structure that drives both mobile apps and website widgets.[4]

2.1.1 Architectural Overview and API Discovery

The architecture of Abfall+ relies on a unique identifier for the waste management agency rather than the end-user. This design choice simplifies the data extraction process for Fessi, as it eliminates the need for individual user accounts. The primary method for discovering the necessary connection parameters involves inspecting the network traffic of a municipality's "Abfuhrtermine" (Collection Dates) webpage. As detailed in technical analyses of the platform, the backend is typically accessed via https://api.abfallplus.de or https://api.abfall.io, with a specific hash key passed as a GET parameter.[4]

For example, in the district of Waldshut, the integration is keyed by a specific hash (5b0384147b5bc055c30fee1fb6db6f76), which grants access to the public data for that region. This key acts as a "tenant ID" in the multi-tenant architecture of Abfall+. Fessi's ingestion layer must be designed to store these keys as configuration variables, mapped to their respective university cities.[4]

2.1.2 Endpoint Structure and Parameter Hierarchy

The data retrieval process for Abfall+ is hierarchical, requiring a specific sequence of calls to resolve a user's location into a collection schedule. This hierarchy must be mirrored in the Fessi knowledge graph's entity resolution logic. The interaction flow is typically strictly ordered:

1. **Municipality Resolution (f_id_kommune):** The system first queries for the ID of the specific town or municipality within the district.
2. **District/Street Resolution (f_id_bezirk, f_id_strasse):** Once the municipality ID is known, a subsequent call retrieves the streets. In some larger administrative zones, an intermediate "District" layer exists.
3. **Waste Type Selection (f_id_abfalltyp):** Finally, the specific bins (e.g., Residual, Bio, Paper) are identified by their internal IDs.

The research identifies the standard payload for a schedule request as a POST request to the /index.php endpoint with the modul parameter set to termine (dates). The body of the request typically follows this structure:

Parameter	Type	Description	Example Value
key	String (Hash)	The unique tenant ID for the waste agency.	5b0384147b5...
f_id_kommune	Integer	The internal ID for the municipality.	270
f_id_strasse	Integer	The internal ID for the street.	15720
f_id_abfalltyp_0	Integer	ID for the first requested bin type.	51 (e.g., Restmüll)
f_id_abfalltyp_1	Integer	ID for the second requested bin type.	17 (e.g., Bio)
modul	String	The action to perform.	termine

Implications for Fessi: The values for f_id_abfalltyp are non-standard. An ID of "51" might represent "Restmüll" in one district and "Altpapier" in another. Therefore, Fessi cannot assume global consistency for these IDs. The ingestion pipeline must include an "Initialization Step" where it calls the API's metadata endpoint to fetch the name-to-ID mapping for each configured city.[4] This mapping is then stored in the knowledge graph to translate the API's numeric response into the semantic ontology (e.g., waste:ResidualWaste).

2.1.3 Community-Driven Configuration Databases

A critical resource for populating Fessi's database of supported cities is the open-source community, specifically the Home Assistant ecosystem. The repository mampfes/hacs_waste_collection_schedule acts as a de facto registry of these parameters. The source code within this repository contains pre-configured tuples of (key, kommune_id, provider_name) for hundreds of German cities.[6]

By parsing the abfall_io.py and associated configuration files from this repository, Fessi can bootstrap its support for dozens of university locations—including Landshut, Potsdam-Mittelmark, and various districts in Baden-Württemberg—without manual reverse-engineering for each one. This significantly reduces the "Time to First Byte" for the project's data acquisition phase.[7]

2.2 The MyMüll.de / JumoMind Ecosystem

The second major pillar of the German waste data infrastructure is the ecosystem powered by JumoMind, commonly known through the "MyMüll.de" app brand. This provider is particularly dominant in Western Germany (North Rhine-Westphalia, Hesse) and presents a more modern, REST-oriented interface compared to the older PHP-based structure of Abfall+.

2.2.1 RESTful Discovery and Dynamic Querying

Unlike Abfall+, which often requires scraping a website to find the API key, the JumoMind architecture allows for dynamic discovery via a central endpoint: https://mymuell.jumomind.com/mmapp/api.php.[8] This endpoint supports a query parameter r (resource) that allows the client to navigate the data tree dynamically.

The relevance of this for Fessi is profound: instead of hardcoding cities, the system can periodically query r=cities to retrieve a complete, up-to-date list of all supported municipalities in the JumoMind network. This allows Fessi to automatically expand its coverage as new cities sign contracts with MyMüll.

2.2.2 The "Area ID" Abstraction

The JumoMind system uses a unified area_id to represent the user's location, which can correspond to a specific street or a sub-district. The workflow for data retrieval is as follows:

1. **City Discovery:** GET...&r=cities returns a list of objects containing id (City ID) and name.
2. **Street Discovery:** GET...&r=streets&city_id=X returns the list of streets for that city, each with an associated area_id.
3. **Schedule Retrieval:** GET...&r=trash&city_id=X&area_id=Y returns the actual collection events.[9]

Data Richness: The response from the trash endpoint is typically a JSON object that includes not just the date, but also metadata such as the color of the bin (e.g., #FF0000 for

Red/Hazardous). This attribute should be ingested into the Fessi Knowledge Graph's visual layer, allowing the chatbot to respond with cues like "Put it in the Red Bin" rather than just "Hazardous Waste," thereby improving user experience.[8]

2.2.3 Supported University Locations

The research snippets highlight several university cities that utilize this backend, explicitly mentioning cities like **Flensburg** (Europa-Universität Flensburg), **Bad Homburg**, and districts around **Gießen** and **Marburg** (via the local waste management unions).[10] The integration of the JumoMind API is therefore essential for covering the central and western German university landscape.

2.3 The Müllmax Legacy Systems

While app-based APIs represent the modern standard, many large, established municipal utilities rely on the "Müllmax" web portal system. This system is less accessible to machine-to-machine communication and represents the "legacy" tier of data sources that Fessi must accommodate.

2.3.1 The Scraping Necessity

Müllmax portals do not typically expose a clean JSON API. Instead, they function as server-side rendered web forms where a user selects their street and the server generates an HTML table or a PDF calendar. For Fessi to ingest this data, it must employ a "Web Scraping" adapter that simulates the POST requests of the web form.[11]

The complexity here lies in the session management and the multi-step selection process often required (e.g., Select District -> Reload Page -> Select Street -> Reload Page -> Select House Number). The hacs_waste_collection_schedule documentation notes that Müllmax integrations often require parsing the HTML response to extract the specific hidden input fields generated by the server for each session state.[12]

2.3.2 ICS Export as a Stabilization Strategy

To mitigate the fragility of HTML scraping, Fessi should prioritize finding the "ICS Export" link often buried in the final stage of the Müllmax workflow. Once a valid ICS (iCalendar) URL is generated for a specific student dormitory or campus building, this URL is often stable for the remainder of the year. The system strategy should be to "scrape once" to get the ICS URL, and then "poll periodically" on that stable URL, rather than traversing the web form for every user query.[6] This reduces the load on the Müllmax servers and increases the reliability of the Fessi chatbot.

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3. The Regulatory Backbone: The

Abfallverzeichnis-Verordnung (AVV)

While collection schedules inform the user *when* to dispose of waste, the core intelligence of Fessi—knowing *what* an item is and *how* to classify it—must be grounded in the legal framework. The **Abfallverzeichnis-Verordnung (AVV)** serves as the "Ontological Truth" for the system. It is the German implementation of the European Waste Catalogue (EWC) and provides the precise, six-digit code required for regulatory compliance.[2]

3.1 Hierarchical Structure of the AVV

The AVV is not merely a flat list; it is a hierarchical taxonomy that groups waste based on its origin and material composition. Fessi's internal ontology should mirror this structure to allow for sophisticated reasoning. The codes are structured as follows:

- **Chapters (01 - 20):** The broad source of the waste. For a university chatbot, **Chapter 20** ("Municipal wastes including separately collected fractions") is the most critical.[3]
- **Sub-Chapters:** These define specific streams within the source. For example, 20 01 covers "Separately collected fractions."
- **Waste Codes:** The specific material definition.

Key codes that must be hardcoded into Fessi's decision logic include:

- **20 01 01:** Paper and cardboard (*Papier und Pappe*).
- **20 01 02:** Glass (*Glas*).
- **20 01 08:** Biodegradable kitchen and canteen waste (*Biologisch abbaubare Küchen- und Kantinenabfälle*).
- **20 01 39:** Plastics (*Kunststoffe*).
- **20 03 01:** Mixed municipal waste (*Gemischte Siedlungsabfälle* / Restmüll).[2]

3.2 Hazardous Waste and Mirror Entries

A critical feature of the AVV is the concept of "Mirror Entries" and the asterisk notation (*) for hazardous waste. Fessi must be able to distinguish between safe and hazardous versions of similar items to protect the university and its students from liability and safety risks.

- **The Asterisk Rule:** Any AVV code suffixed with an asterisk (e.g., 20 01 21* for Fluorescent tubes) is legally classified as hazardous (*gefährlich*). Fessi must trigger a "Hazard Alert" in the UI when such an item is identified, explicitly blocking the user from suggesting standard bin disposal.[2]
- **Mirror Entries:** Some waste types have two entries depending on their contamination. For example, wood can be 20 01 37* (wood containing hazardous substances) or 20 01 38 (wood other than that mentioned in 20 01 37).
    - *Chatbot Logic:* If a user asks about "Wood from the chemistry lab renovations," Fessi must ask a clarifying question: "Was this wood treated with preservatives or chemicals?" to determine the correct AVV path.[3]

3.3 Data Source Integration for AVV

Unlike the APIs for schedules, the AVV is a static dataset. It is available as a legal text PDF from the Federal Ministry of Justice [3], but for computational use, Fessi requires structured data.

- **Destatis & Environmental Agencies:** Statistical offices often publish the AVV list as Excel (.xlsx) or CSV files for reporting purposes.[13]
- **International Databases:** Since the EWC is a European standard, English-language databases like wikiwaste.org.uk or csg.co.uk can be used to bootstrap the code list, which can then be joined with the German descriptions from the official AVV text.[15]

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4. The Consumer Layer: The "Abfall-ABC" and Semantic Normalization

While the AVV provides the legal code, students do not search for "AVV 20 03 01"; they search for "Pizza box" or "broken toaster." The mapping between these natural language terms and the disposal categories constitutes the "Abfall-ABC" (Waste Dictionary). This is the semantic layer of Fessi.

4.1 The Berlin BSR Dataset: The Gold Standard

The research identifies the **Berliner Stadtreinigung (BSR)** as the premier source for this semantic data. Unlike most municipalities that lock this data inside search widgets, Berlin publishes its full "Abfall-ABC" and "Recyclinghöfe" data as high-quality JSON files on its Open Data portal.[17]

- **Data Richness:** The BSR dataset contains thousands of common items (e.g., "Aktenordner" - File Folder) mapped to disposal routes. It also includes "Avoidance Tips" (*Vermeidungstipps*) and "Recycling Tips" (*Verwertungstipps*).
- **Strategy:** Fessi should use the BSR dataset as its "Base Ontology." Even though disposal rules vary by city, the *list of items* is universal. A "File Folder" is a physical object that exists in Munich just as it does in Berlin. By importing the BSR item list, Fessi immediately gains a vocabulary of 2,000+ items.
- **Normalization:** Fessi can then treat local rules as "overrides" to this base list. If Munich has a different rule for file folders, only that specific relationship needs to be updated in the graph, while the item definition and synonyms remain derived from the Berlin data.[17]

4.2 Handling Regional Variations and "Sorting Myths"

A major challenge identified in the research is the conflict in sorting rules between districts. The "Abfall-ABC" is not static across Germany.

4.2.1 The Pizza Box Paradox

One of the most frequent points of confusion is the disposal of pizza boxes.

- **Variation:** In some jurisdictions, if a pizza box is even slightly greasy, it is strictly classified as *Restmüll* because the grease ruins the paper recycling pulp. In others, empty boxes are accepted in the paper bin.[18]
- **Data Source:** This nuanced information is rarely in the API. It is found in the "Sortierhilfen" (Sorting Help) PDFs and FAQs on municipal websites.
- **Integration:** Fessi requires a "Condition Attribute" in its knowledge graph. The node for "Pizza Box" should not point directly to a bin but to a decision node: "Is it greasy?" -> Yes -> Restmüll; No -> Altpapier.

4.2.2 Bioplastics in the Bio-Bin

Another critical area of divergence is "compostable plastics" (Bioplastik).

- **The Consensus:** Most German waste processors (including those in Heilbronn and Munich) actively ban these bags from the Biotonne because their composting cycle is too slow for industrial plants.[20]
- **Fessi Logic:** The default rule in Fessi must be to *reject* bioplastics from the organic bin unless a specific positive confirmation is found for the user's city. This "fail-safe" approach prevents contamination of the compost stream.

4.3 Scraping Proprietary ABC Widgets

For cities that do not offer Open Data like Berlin, the Abfall-ABC is hidden behind AJAX widgets on sites powered by Abfall+.

- **Technique:** Reverse-engineering of these widgets reveals that they often load the data via a GET request to endpoints like api.abfall.io/customers/.../widgets/abfall-abc.html.[22]
- **Extraction:** The response is often an HTML fragment containing <table> rows with class names like abfallart (Waste Type) and entsorgung (Disposal). Fessi can employ a specialized crawler to extract these pairs (e.g., "Acrylglas" -> "Sperrmüllcontainer") to build city-specific knowledge graphs.[22]

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5. Geospatial Infrastructure: Wertstoffhöfe and Hazardous Waste

For waste streams that cannot be handled curbside (bulky waste, chemicals, large electronics), Fessi must guide the user to a physical location. This requires a robust geospatial data strategy.

5.1 Wertstoffhöfe (Recycling Centers)

The "Wertstoffhof" is the central hub for special disposal.

- **Munich (AWM) Open Data:** Munich provides a best-in-class dataset for this. The awm_wertstoffhoefe_opendata set is available via WFS (Web Feature Service) and GeoJSON. It includes precise coordinates, opening hours, and lists of accepted fractions.[23]
- **Berlin (BSR):** Similarly, Berlin provides a Standorte (Locations) JSON file.[17]
- **OpenStreetMap (OSM) for the "Long Tail":** For university towns without such portals (e.g., Heidelberg, Göttingen), OpenStreetMap is the most reliable source. The community maintains high-quality data under the amenity=recycling and recycling_type=centre tags.[24]
    - **Attributes:** Fessi should query OSM for tags like opening_hours, recycling:wood, recycling:batteries, and recycling:electronics to determine the capabilities of each center.

5.2 The "Schadstoffmobil" (Hazardous Waste Mobile)

The most volatile data source is the schedule for the mobile hazardous waste collection vehicle.

- **The Challenge:** Unlike fixed centers, these move daily. The data is often locked in PDF calendars.[25]
- **Aachen Exception:** The city of Aachen is a rare positive exception, publishing a GeoJSON dataset specifically for Termine Schadstoffmobil.[27] This allows for direct map integration.
- **General Strategy:** For other cities, Fessi must rely on the Abfall+ or JumoMind APIs, which sometimes treat the Schadstoffmobil stops as "events" in the calendar feed (modul=termine). If this fails, PDF scraping of the "Umweltkalender" is the only fallback.
- **Rules of Acceptance:** Fessi must strictly inform users of the acceptance limits found in the research (e.g., max 20kg, max 20L containers, no mixed chemicals) to prevent students from being turned away at the mobile stop.[25]

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6. Architectural Design: The "Fessi" Knowledge Graph

Based on the discovered sources, the recommended architecture for Fessi is a "Federated Knowledge Graph" that acts as a meta-layer over the fragmented sources.

6.1 The Ingestion Pipeline (The Wrapper Pattern)

Fessi should not be a monolith but a collection of "Source Adapters."

- **Adapter A (SaaS):** Connects to Abfall+ and JumoMind APIs. Uses the hacs_waste_collection_schedule configurations to instantiate the correct client for a given ZIP code.

- **Adapter B (Open Data):** Ingests the BSR JSON and AWM GeoJSON to populate the semantic and geospatial base layers.
- **Adapter C (Regulatory):** Loads the AVV codes to ensure legal compliance in classification.

6.2 The Knowledge Graph Schema

The graph should consist of the following primary entities:

- **UserLocation**: Defined by City ID and Street ID (mapped from API).
- **WasteObject**: (e.g., "Pizza Box"). Has properties synonyms (from SKOS) and avv_code.
- **DisposalRule**: Connects a WasteObject to a DisposalMethod (e.g., "Blue Bin") valid for a specific UserLocation.
- **CollectionEvent**: Dynamic entity generated from the API schedule (Date + DisposalMethod).

6.3 Table: Recommended Data Source Mapping for Fessi

The following table summarizes the precise data sources Fessi should prioritize for different categories of information.

Data Category	Primary Source	Backup Source	Format	Key Utility
Collection Schedules	**Abfall+ API** [4] / **JumoMind API** [8]	**Müllmax ICS** [12]	JSON / ICS	Dynamic pickup dates (Blue/Black/Yellow bins).
Waste Taxonomy (Base)	**Berlin BSR Open Data** [17]	**Abfall+ Widget Scraping** [22]	JSON	Baseline vocabulary of 2,000+ items.
Legal Classification	**AVV / EWC Lists** [2]	**Destatis Reports** [13]	PDF / CSV	Regulatory codes (e.g., 20 03 01) for compliance.
Recycling Centers	**Munich/Berlin Open Data** [23]	**OpenStreetMap** [24]	GeoJSON / WFS	GPS coordinates and opening

				hours.
Hazardous Waste Rules	**Municipal PDFs** [25]	**Federal Environment Agency**	PDF / HTML	Strict acceptance limits (kg/L) for safety.
City Configuration	**Home Assistant Repo** [6]	Manual Discovery	Python Code	API Keys and City IDs for hundreds of towns.

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7. Conclusion

The digital landscape of German waste management, while administratively fragmented, presents a highly structured technical opportunity for the "Fessi" project. The convergence of municipal IT onto a few key SaaS platforms (Abfall+, JumoMind) allows Fessi to achieve broad geographic coverage through a limited number of API integrations. By combining these dynamic schedule sources with the semantic richness of Berlin's Open Data and the regulatory precision of the AVV, Fessi can offer a user experience that is far superior to the disjointed ecosystem currently facing university students.

The path forward requires a focus on robust API wrapping—leveraging the groundwork laid by the open-source community—and a sophisticated ontology that can handle the subtle regional variations in sorting rules. With this architecture, Fessi will not only tell a student *when* to put out the bin but *what* exactly belongs in it, grounded in both local reality and federal law.

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